

Erdős Problem 920 from an Off-Diagonal Ramsey Bound

A complete proof, quantitative consequences, and a research appendix on the remaining logarithmic gap

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Abstract

For fixed $k \geq 3$, let $f_k(n)$ be the largest chromatic number of an n -vertex graph containing no copy of K_k . Erdős Problem 920 asks whether, for every fixed $k \geq 4$, there is a constant $c_k > 0$ such that

$$f_k(n) \gg_k \frac{n^{1-1/(k-1)}}{(\log n)^{c_k}}.$$

Bradač's 2026 lower bound for off-diagonal Ramsey numbers,

$$r(k, t) \gg_k \frac{t^{k-1}}{(\log t)^{2k-4}},$$

settles the problem after an explicit asymptotic inversion. We give a quantifier-complete proof valid for every sufficiently large integer order n , obtaining

$$f_k(n) \gg_k \frac{n^{1-1/(k-1)}}{(\log n)^{2(k-2)/(k-1)}}.$$

We also derive the complementary chromatic upper estimate from the Ajtai–Komlós–Szemerédi Ramsey bound, formulate the equivalent minimum-order problem, and record the precise remaining polylogarithmic gap.

The appendices collect a longer research program aimed at the logarithmically optimal endpoint. Every item is explicitly labelled as proved, refuted, conditional, experimental, or open, so that exploratory finite-geometric calculations are not confused with the complete solution of the original problem.

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1 Statement, scope, and priority

For integers $k, n \geq 1$, define

$$f_k(n) := \max\{\chi(G) : |V(G)| = n \text{ and } K_k \not\subseteq G\}.$$

All graphs in the main theorem are finite and simple, and all logarithms are natural.

Problem 1.1 (Erdős Problem 920). For every fixed $k \geq 4$, is there a constant $c_k > 0$ such that

$$f_k(n) \gg_k \frac{n^{1-1/(k-1)}}{(\log n)^{c_k}}?$$

The deep input is Bradač's theorem on off-diagonal Ramsey numbers. The deduction from that theorem to Problem 920 is elementary, but the quantifier order and logarithmic inversion are worth recording carefully.

A short manuscript by Miroslav Lžičar, submitted shortly before the present exposition was completed, independently records essentially the same Ramsey-to-chromatic corollary. Accordingly, this document makes no priority claim for that deduction. Its purpose is to present a self-contained expanded proof and to preserve, with an explicit trust boundary, the subsequent investigation of the sharper logarithmic endpoint.

2 Ramsey numbers and the finite transfer

Let $r(k, t)$ be the least integer N such that every graph on N vertices contains either a clique of order k or an independent set of order t .

Lemma 2.1 (Finite Ramsey-to-chromatic transfer). *Let k, t, n be positive integers with $t \geq 2$. If*

$$r(k, t) > n,$$

then

$$f_k(n) \geq \left\lceil \frac{n}{t-1} \right\rceil.$$

Proof. Because $n < r(k, t)$, there is an n -vertex graph G containing neither a K_k nor an independent set of order t . Thus

$$\alpha(G) \leq t-1.$$

Every color class of a proper coloring is independent. Hence, if $\chi(G) = q$, then

$$n \leq q\alpha(G) \leq q(t-1),$$

so

$$q \geq \left\lceil \frac{n}{t-1} \right\rceil.$$

Since G is K_k -free, the same lower bound holds for $f_k(n)$. □

The next theorem is the general inversion mechanism.

Theorem 2.2 (General transfer theorem). *Fix $k \geq 3$. Suppose there are constants*

$$a > 0, \quad d > 1, \quad A \geq 0,$$

and $t_0 \geq 2$ such that

$$r(k, t) \geq a \frac{t^d}{(\log t)^A} \quad (t \geq t_0).$$

Then there are constants $C > 0$ and $N \in \mathbb{N}$ such that

$$f_k(n) \geq C \frac{n^{1-1/d}}{(\log n)^{A/d}} \quad (n \geq N).$$

Proof. Choose $B \geq 1$ so large that

$$aB^d > 1.$$

For $n \geq 3$, set

$$x_n := Bn^{1/d}(\log n)^{A/d}, \quad t_n := \lceil x_n \rceil.$$

Then $t_n \rightarrow \infty$, so $t_n \geq t_0$ eventually.

We also have $t_n \leq n$ eventually. Indeed, once $x_n \geq 1$,

$$t_n \leq x_n + 1 \leq 2x_n,$$

and therefore

$$\log \frac{t_n}{n} \leq \log(2B) - \left(1 - \frac{1}{d}\right) \log n + \frac{A}{d} \log \log n \rightarrow -\infty.$$

Thus, for all sufficiently large n ,

$$\log t_n \leq \log n.$$

The assumed Ramsey bound gives

$$\begin{aligned} r(k, t_n) &\geq a \frac{t_n^d}{(\log t_n)^A} \\ &\geq a \frac{x_n^d}{(\log n)^A} \\ &= aB^d n \\ &> n. \end{aligned}$$

By Lemma 2.1,

$$f_k(n) \geq \frac{n}{t_n - 1}.$$

Since $\lceil x \rceil - 1 < x$ for every real x ,

$$t_n - 1 < x_n = Bn^{1/d}(\log n)^{A/d}.$$

Consequently,

$$f_k(n) > \frac{1}{B} \frac{n^{1-1/d}}{(\log n)^{A/d}}.$$

Take $C = 1/B$ after increasing N to cover all preceding eventual conditions. $\square$

Remark 2.3 (Exact orders). The argument produces a graph on exactly n vertices for every sufficiently large integer n . No sparse sequence of admissible orders and no unspoken padding argument are needed.

3 The complete solution of Problem 920

We use the following theorem in its current strengthened form.

Theorem 3.1 (Bradač, 2026). *For every fixed integer $k \geq 3$, there is a constant $a_k > 0$ such that*

$$r(k, t) \geq a_k \frac{t^{k-1}}{(\log t)^{2k-4}}$$

for every sufficiently large integer t .

Theorem 3.2 (Affirmative resolution of Erdős Problem 920). *For every fixed integer $k \geq 3$, there are constants $C_k > 0$ and $N_k \in \mathbb{N}$ such that*

$$f_k(n) \geq C_k \frac{n^{1-1/(k-1)}}{(\log n)^{2(k-2)/(k-1)}} \quad (n \geq N_k).$$

In particular, for every $k \geq 4$, Problem 920 has an affirmative answer, and one may take

$$c_k = \frac{2(k-2)}{k-1}.$$

Proof. Apply Theorem 2.2 to Theorem 3.1 with

$$d = k - 1, \quad A = 2k - 4.$$

Then

$$1 - \frac{1}{d} = 1 - \frac{1}{k-1}$$

and

$$\frac{A}{d} = \frac{2k-4}{k-1} = \frac{2(k-2)}{k-1}.$$

This is exactly the asserted inequality. For $k \geq 4$, the logarithmic exponent is positive. $\square$

Remark 3.3 (The case $k = 4$). The theorem gives

$$f_4(n) \gg \frac{n^{2/3}}{(\log n)^{4/3}},$$

which is the same logarithmic exponent obtained from the Mattheus–Verstraete lower bound for $r(4, t)$.

Remark 3.4 (Why the version matters). An earlier version of Bradač’s work had polynomial exponent $k - 2$ in the Ramsey lower bound. The transfer theorem would then yield the exponent

$$1 - \frac{1}{k-2},$$

which is strictly smaller than the exponent requested in Problem 920 when $k \geq 5$. The strengthened exponent $k - 1$ is exactly what closes the problem.

4 The current logarithmic frontier

Put

$$a_k := \frac{k-2}{k-1}.$$

Theorem 3.2 gives

$$f_k(n) \gg_k \frac{n^{a_k}}{(\log n)^{2a_k}}.$$

The classical upper bound of Ajtai, Komlós, and Szemerédi states that, for fixed $k \geq 3$,

$$r(k, t) \ll_k \frac{t^{k-1}}{(\log t)^{k-2}}.$$

This implies a complementary chromatic upper bound without the historical $\log \log n$ loss.

Proposition 4.1 (Hereditary independent-set extraction). *For each fixed $k \geq 3$, there is $c_k > 0$ such that every sufficiently large m -vertex K_k -free graph H satisfies*

$$\alpha(H) \geq c_k m^{1/(k-1)} (\log m)^{(k-2)/(k-1)}.$$

Proof. Choose

$$t = \left\lfloor \delta_k m^{1/(k-1)} (\log m)^{(k-2)/(k-1)} \right\rfloor$$

with $\delta_k > 0$ sufficiently small. Then

$$\frac{t^{k-1}}{(\log t)^{k-2}} = \left(\delta_k^{k-1} (k-1)^{k-2} + o(1) \right) m.$$

For sufficiently small δ_k , the Ramsey upper bound gives

$$r(k, t) \leq m.$$

Since H contains no K_k , it must contain an independent set of order t . □

Theorem 4.2 (Chromatic upper bound). *For every fixed $k \geq 3$,*

$$f_k(n) \ll_k \frac{n^{a_k}}{(\log n)^{a_k}}.$$

Proof. Let G be an n -vertex K_k -free graph. Repeatedly remove an independent set supplied by Proposition 4.1, assigning one new color to each removed set.

When the residual order m satisfies

$$2^{j-1} < m \leq 2^j,$$

every removed independent set has order at least

$$c_k 2^{(j-1)/(k-1)} j^{a_k}.$$

Thus the number of colors used while the residual order lies in this dyadic interval is at most

$$C_k \frac{2^j}{2^{j/(k-1)} j^{a_k}} = C_k \frac{2^{ja_k}}{j^{a_k}}.$$

If $J = \lceil \log_2 n \rceil$, then

$$\chi(G) \leq C_k \sum_{j \leq J} \frac{2^{ja_k}}{j^{a_k}} + O_k(1).$$

Because $a_k > 0$, the sum is dominated by its final terms, giving

$$\chi(G) \ll_k \frac{2^{Ja_k}}{J^{a_k}} \asymp_k \frac{n^{a_k}}{(\log n)^{a_k}}.$$

Maximizing over all admissible G proves the result. □

Combining Theorems 3.2 and 4.2,

$$\boxed{\frac{n^{a_k}}{(\log n)^{2a_k}} \ll_k f_k(n) \ll_k \frac{n^{a_k}}{(\log n)^{a_k}}}.$$

In particular,

$$f_k(n) = n^{a_k + o(1)}.$$

The polynomial exponent is settled. The remaining problem is purely polylogarithmic.

5 Equivalent minimum-order formulation

Define

$$h_k(q) := \min\{|V(G)| : K_k \not\subseteq G \text{ and } \chi(G) \geq q\}.$$

The preceding estimates invert to

$$\boxed{q^{(k-1)/(k-2)} \log q \ll_k h_k(q) \ll_k q^{(k-1)/(k-2)} (\log q)^2}.$$

Indeed, the upper bound for f_k implies the lower bound for h_k , while Theorem 3.2 implies the upper bound for h_k by choosing

$$n \asymp_k q^{(k-1)/(k-2)} (\log q)^2.$$

Thus the sharper endpoint problem is equivalent to removing one logarithm from the upper bound for $h_k(q)$.

6 Trust boundary

The logical status of the main document is as follows.

- The finite transfer, asymptotic inversion, hereditary extraction, dyadic coloring argument, and minimum-order inversion are proved here.
- The deep lower bound in Theorem 3.1 is imported from Bradač's preprint.
- The off-diagonal Ramsey upper bound used in Proposition 4.1 is imported from Ajtai–Komlós–Szemerédi.
- None of the exploratory appendices below is needed for Theorem 3.2.
- Exploratory claims are assigned an explicit status and are not presented as established literature unless a proof is included.

A Research appendix: the logarithmically optimal endpoint

The remaining target is

$$f_k(n) \stackrel{?}{\gg}_k \left(\frac{n}{\log n} \right)^{a_k},$$

or equivalently

$$h_k(q) \stackrel{?}{\ll}_k q^{(k-1)/(k-2)} \log q.$$

For $k = 4$, this asks for a K_4 -free graph on $O(q^3 \log q)$ vertices with chromatic number $\Omega(q^2)$.

The remaining appendices record a verification-controlled search for this stronger statement. They are not part of the solution of Problem 920.

A.1 Status ledger

Item	Status	Meaning
Ramsey-to-chromatic transfer	PROVED	Complete and used in the main theorem.
Hereditary AKS coloring bound	PROVED	Gives the upper endpoint $n^{a_k}/(\log n)^{a_k}$.
Additive resource packing	REFUTED	The initially proposed inequality has the wrong sign and gives an upper bound on the number of classes.
Quadratic collision certificate	PROVED	A bound on $\sum_j I_j ^2$ would imply the desired chromatic lower bound.
Laminar zero-rectangle identity	PROVED	Rewrites the quadratic certificate as an area-packing problem.
Support-exact low-rank matrix	PROVED	Encodes the arc set of the finite-geometric digraph by a matrix of rank $O_t(q^t)$.
Constant conflict refinement	REFUTED	Shift graphs occur inside acyclic sets, forcing an $\Omega(\log q)$ refinement loss.
Separate-container counting	REFUTED	Its innovation entropy is one logarithm too large to prove the sharp endpoint.
Regular-spread phase model for $k = 4$	CONDITIONAL	Algebraically derived candidate construction; not used in the main theorem and not independently literature-verified.
Universal quadratic gain lower tail	REFUTED	Two-row palette splitting gives probability $\exp(-\Theta(m))$, not $\exp(-\Theta(m^2/q))$.
Single balanced-block phase lemma	REFUTED	A union of skew balanced lines defeats the proposed dichotomy.
Multi-block phase-energy control	OPEN	The final obstruction in the exploratory $k = 4$ route.

B Why a one-class Ramsey construction loses a logarithm

Bradač's construction is based on a directed graph $D^*(t, q)$ built from projective incidence data. Its vertices are incident projective pairs (a, b) , and the asymmetric arc relation has the form

$$(a, b) \longrightarrow (a', b') \iff a \perp b' \text{ and } a' \not\perp b.$$

The digraph contains no transitive tournament of order $t + 1$. Retaining only arcs that point forward in a chosen order therefore produces a K_{t+1} -free graph.

The independent-tuple argument admits a marked/unmarked tree description with the following scales:

$$M \asymp q^{2t-1}, \quad h = O_t(q^t), \quad w = O_t(q \log q).$$

Here M is the ambient number of vertices, h bounds the marked branching, and every root-to-leaf path has at most w unmarked transitions.

The exceptional unmarked choices cost roughly

$$w \log \frac{M}{h} \asymp q \log q \cdot (t - 1) \log q = \Theta_t(q(\log q)^2)$$

for one color class. At the desired chromatic scale there are

$$r \asymp q^{t-1}$$

classes, so treating their exceptional budgets independently costs

$$\Theta_t(q^t(\log q)^2).$$

The available vertex budget is only

$$N \asymp q^t \log q.$$

This is the exact missing logarithm in the separate-container proof.

C A sign correction for resource packing

An early proposal sought a resource space Ω and weights $w_I(\omega)$ satisfying

$$\sum_{\omega \in \Omega} w_I(\omega) \geq L$$

for every independent set I , together with

$$\sum_j w_{I_j}(\omega) \leq C$$

for every resource ω across a coloring. Summing gives

$$rL \leq C|\Omega|,$$

and hence

$$r \leq \frac{C|\Omega|}{L}.$$

This is an upper bound on the number of classes, not the required lower bound.

Any valid resource method must instead bound the amount of the vertex set coverable by a small family of classes, or use a convex certificate whose direction is compatible with a lower bound for the number of colors.

D The quadratic collision certificate

Suppose a candidate graph has

$$N \asymp q^t \log q$$

and every partition into independent sets $I_1, \dots, I_r$ satisfies

$$\sum_{j=1}^r |I_j|^2 \leq C_t q^{t+1} (\log q)^2.$$

Then Cauchy–Schwarz gives

$$N^2 = \left(\sum_j |I_j| \right)^2 \leq r \sum_j |I_j|^2,$$

so

$$r \geq \frac{N^2}{C_t q^{t+1} (\log q)^2} \asymp_t q^{t-1}.$$

Thus a quadratic or collision-type certificate avoids the generic logarithmic integrality gap of ordinary fractional coloring.

E Laminar zero rectangles

Let an ordered independent or acyclic class be

$$I = (v_1, \dots, v_s).$$

Recursively split every interval into two consecutive halves. At each internal node ν , write X_ν for the left half and Y_ν for the right half. Independence in the forward graph implies that there are no directed arcs from X_ν to Y_ν .

Lemma E.1 (Laminar area identity). *For every ordered class I ,*

$$\sum_{\nu \in \mathcal{T}(I)} |X_\nu| |Y_\nu| = \binom{|I|}{2}.$$

Proof. Every ordered pair v_i, v_j with $i < j$ is separated for the first time at a unique lowest internal node of the binary splitting tree. It contributes exactly once to the left-right product at that node. $\square$

Summing over the color classes gives

$$\sum_j |I_j|^2 = N + 2 \sum_\nu |X_\nu| |Y_\nu|.$$

The desired quadratic certificate is therefore equivalent to a Carleson-type packing inequality for a vertex-disjoint laminar forest of directed zero rectangles.

F A support-exact low-rank matrix

Let the ambient projective vector dimension be

$$d = t + 1.$$

Choose normalized vector representatives for the projective points. For flags

$$x = (a_x, b_x), \quad y = (a_y, b_y),$$

define over $\mathbb{F}_q$

$$Q_{xy} := \langle a_y, b_x \rangle \left(1 - \langle a_x, b_y \rangle^{q-1} \right).$$

Proposition F.1 (Arc support).

$$Q_{xy} \neq 0 \iff x \longrightarrow y.$$

Proof. For $z \in \mathbb{F}_q$,

$$1 - z^{q-1} = \begin{cases} 1, & z = 0, \\ 0, & z \neq 0. \end{cases}$$

Thus the second factor is nonzero exactly when $a_x \perp b_y$, while the first factor is nonzero exactly when $a_y \not\perp b_x$. $\square$

Proposition F.2 (Rank bound).

$$\text{rank}_{\mathbb{F}_q} Q \leq d \left(1 + \binom{q+d-2}{d-1} \right) = O_t(q^t).$$

Proof. The term $\langle a_y, b_x \rangle$ has rank at most d . Expanding

$$\langle a_x, b_y \rangle^{q-1} = \sum_{|\alpha|=q-1} c_\alpha a_x^\alpha b_y^\alpha$$

shows that the second term in

$$Q_{xy} = \langle a_y, b_x \rangle - \langle a_y, b_x \rangle \langle a_x, b_y \rangle^{q-1}$$

is a sum of at most

$$d \binom{q+d-2}{d-1}$$

rank-one matrices. $\square$

If

$$Q = UV^\top$$

and I is acyclic, then after a topological ordering the principal block $Q[I, I]$ is strictly triangular. Hence it is nilpotent.

The nonzero eigenvalues of $U_I V_I^\top$ and $V_I^\top U_I$ agree. Therefore

$$T_I := V_I^\top U_I = \sum_{x \in I} v_x u_x^\top$$

is nilpotent.

This converts a coloring into a partition of structured rank-one atoms into nilpotent sums. The global rank bound alone is too weak, but the factorization suggests a rank-to-projective-structure route.

G A shift-graph obstruction

A natural shortcut would refine every acyclic class into a bounded number of conflict-independent pieces. This is impossible.

Let

$$P_1, \dots, P_m$$

be points in a projective plane with no three collinear. For $i < j$, let L_{ij} be the line through P_i, P_j , and define the flag

$$v_{ij} := (P_j, L_{ij}).$$

The asymmetric incidence relation gives

$$v_{ij} \longrightarrow v_{j\ell} \quad (i < j < \ell),$$

and no other arcs of this form.

The underlying conflict graph is the shift graph S_m , whose chromatic number is

$$\chi(S_m) = \lceil \log_2 m \rceil.$$

The terminal index increases along every arc, so the digraph induced by these flags is acyclic.

Consequently an acyclic class may require

$$\Omega(\log q)$$

conflict-independent pieces. Any method based on bounded conflict refinement reproduces the logarithm it seeks to remove.

H Exploratory $k = 4$ regular-spread phase model

Status: conditional. The following algebraic model was derived during the search and used to formulate falsifiable subproblems. It is not needed for the main theorem and is not claimed here as an independently published construction.

Let

$$E = \mathbb{F}_{q^2}, \quad V = E^2$$

viewed as a four-dimensional $\mathbb{F}_q$ -space. Fix a nondegenerate alternating $\mathbb{F}_q$ -bilinear form ω on E . For $a \in E$, define

$$\beta_a((x_1, x_2), (y_1, y_2)) := \omega(x_2 - ax_1, y_2 - ay_1).$$

Its radical is

$$K_a = \{(x, ax) : x \in E\}.$$

Together with

$$K_\infty = \{(0, x) : x \in E\},$$

the projective lines $\mathbb{P}(K_a)$ form a regular spread of $\text{PG}(3, q)$.

Splitting the parameter line into layer labels and kernel labels removes the previously observed singular degree anomalies. After quotienting

$$H := E^*/\mathbb{F}_q^*, \quad |H| = q + 1,$$

each pair of cells is joined either by a perfect matching on H or by no arcs. The inactive case is governed by a subfield-valued cross ratio. This regularization turns the lower-tail problem into a gain-graph coloring problem.

I A universal gain lower tail is false

Status: refuted. A proposed statement asserted that every cell set F with $g(F)$ active pairs should satisfy

$$\Pr(F \text{ is independent}) \leq \exp\left(-c \frac{g(F)}{q}\right).$$

Two rows disprove this.

For two layer parameters $a < b$, the active gain equations can be gauged into equalities

$$X_c = Y_d,$$

where all X_c and Y_d are independent and uniform in a set of order $q + 1$.

Partition the label set into two large disjoint palettes H_0, H_1 . The event

$$X_c \in H_0 \text{ for all } c, \quad Y_d \in H_1 \text{ for all } d$$

forces independence and has probability

$$\exp(-O(m))$$

for a set of order m .

Yet the active-pair count may be

$$\Theta(m^2).$$

At $m \asymp q \log q$, the proposed bound would be

$$\exp(-\Theta(q(\log q)^2)),$$

contradicting the palette event

$$\exp(-O(q \log q)).$$

Thus balanced phases must be extracted and counted by palette entropy before any cluster estimate is applied to the residual frustrated constraints.

J Why a single balanced block is insufficient

Status: refuted. A higher-order proposal asserted that either one balanced phase block contains a fixed fraction of a source set, or the number of high-order phase collisions is $O_r(m^r)$. A union of several pairwise skew balanced lines disproves this.

In the projective interpretation, a balanced phase block is a non-spread projective line. An ordered r -tuple of distinct points on such a line has exactly

$$q + 1 - r$$

common phase bins.

Taking $r + 1$ pairwise skew balanced lines gives a set S in which no line contains $|S|/r$ points, while the total number of r -fold phase collisions is

$$\Theta_r(q|S|^r).$$

The correct moment estimate must therefore retain the contribution of every balanced line, not merely extract one dominant block.

K A multi-block phase-moment identity

Status: conditional within the phase model. Let S be the spread, let S be a set of source points, and for every non-spread projective line ℓ write

$$s_\ell := |S \cap \ell|.$$

Let

$$(x)_r = x(x-1) \cdots (x-r+1).$$

Classifying an ordered r -tuple by its projective span gives

$$M_r(S) \leq (|S|)_r + (q-r) \sum_{\ell \notin S} (s_\ell)_r,$$

where $M_r(S)$ counts common phase bins of ordered distinct r -tuples.

Since every ordered pair lies on a unique line,

$$\sum_{\ell \notin S} (s_\ell)_2 \leq (|S|)_2.$$

If

$$\Delta(S) := \max_{\ell \notin S} s_\ell,$$

then

$$\sum_{\ell \notin S} (s_\ell)_r \leq \Delta(S)^{r-2} (|S|)_2,$$

and hence

$$M_r(S) \leq |S|^r + q\Delta(S)^{r-2}|S|^2.$$

The remaining endpoint obstruction is therefore a multi-block packing and palette problem for the line-energy

$$q \sum_{\ell \notin S} (s_\ell)_r.$$

A single-block stability theorem cannot control it.

L Computational observations

Status: experimental. Finite computations were used only to falsify overly rigid conjectures and to guide the structural search. They do not prove asymptotic statements.

Experiment	Observation
Exact independent-set MILP for small q	Large independent sets need not share one coordinate or lie in an obvious proper subspace.
Exact small colorings	Medium-sized color classes can contain substantial unmarked behavior; “mostly marked” is not automatic at threshold scale.
Coordinate-regular acyclic search	Bounding first- or second-coordinate multiplicities alone does not bound acyclic length.
Spectral diagnostics	Basic Hoffman-type bounds were far below observed greedy chromatic numbers, suggesting that adjacency spectrum alone does not detect the partition obstruction.
Zero-rectangle MILP	Large directed zero rectangles exist, confirming that arbitrary rectangle bounds are false without internal acyclicity or laminarity.
Marked-tail beam search	The one-class Gaussian suppression is visible, but independent color routing restores the missing logarithm.
Phase-design calculations	Pair collisions are highly regular, but balanced cycles create palette phases whose probability scale is only exponential in the number of vertices.

M The exact open sharpening

The original problem is solved. The unresolved strengthening is to determine the true logarithmic exponent in

$$f_k(n) = n^{a_k} (\log n)^{-\beta_k + o(1)}.$$

The established estimates place any such exponent between

$$a_k \quad \text{and} \quad 2a_k.$$

For $k = 4$, the exploratory finite-geometric route reduces the desired endpoint to controlling many mutually compatible medium-sized independent classes. The surviving obstruction is not the maximum size of one independent set. It is the multi-block palette entropy of a partition.

A successful continuation must therefore prove one of the following genuinely stronger mechanisms:

1. a multi-block phase-packing theorem controlling the aggregate line-energy across disjoint classes;
2. a higher-order convex or semidefinite certificate that sees disjointness between independent sets;
3. a modified construction in which exceptional choices are shared by design rather than analyzed after the fact;
4. a different algebraic host whose acyclic or independent partitions admit a direct quadratic collision bound.

N Formalization note

The elementary proof has a short formal structure:

$$n < r(k, t) \implies \exists G, \quad |V(G)| = n, \quad K_k \not\subseteq G, \quad \alpha(G) < t,$$

followed by

$$n \leq \chi(G)\alpha(G) < \chi(G)t.$$

A Lean formalization can therefore isolate Bradač's Ramsey theorem as the only deep imported axiom and kernel-check the entire transfer.

O Conclusion

Bradač's off-diagonal Ramsey theorem gives, for every fixed $k \geq 3$,

$$r(k, t) \gg_k \frac{t^{k-1}}{(\log t)^{2k-4}}.$$

The direct transfer proved here yields

$$f_k(n) \gg_k \frac{n^{1-1/(k-1)}}{(\log n)^{2(k-2)/(k-1)}}.$$

Thus Erdős Problem 920 is resolved affirmatively with

$$c_k = \frac{2(k-2)}{k-1}.$$

The remaining research problem is not the yes-or-no question. It is the sharper determination of the logarithmic exponent, currently trapped between

$$a_k \quad \text{and} \quad 2a_k.$$

References

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